

4.0 Summative Assessment: Network Infrastructure Design Proposal

Introduction

As the IT consultant, I was tasked with designing a secure, reliable, and scalable network infrastructure for a rapidly growing organization. The company operates a headquarters in a large office building and three branch offices located in different cities, with plans to expand further over the next several years. The proposed design must provide efficient communication, centralized resource management, secure remote access for executives, and reliable connectivity between all locations.

This proposal recommends appropriate network topologies, network architecture, network types, IPv4 and IPv6 addressing, network services, an implementation plan, and layered security controls to support current business operations while providing a foundation for future growth.

Network Topology Recommendation

Selecting the appropriate network topology is essential for providing reliability, scalability, and efficient network performance.

Headquarters

A **star topology** is recommended for the **headquarters**. All workstations, printers, wireless access points, and servers connect to managed access switches, which connect to redundant core switches. This design simplifies troubleshooting, isolates individual device failures, and allows the network to expand with minimal disruption.

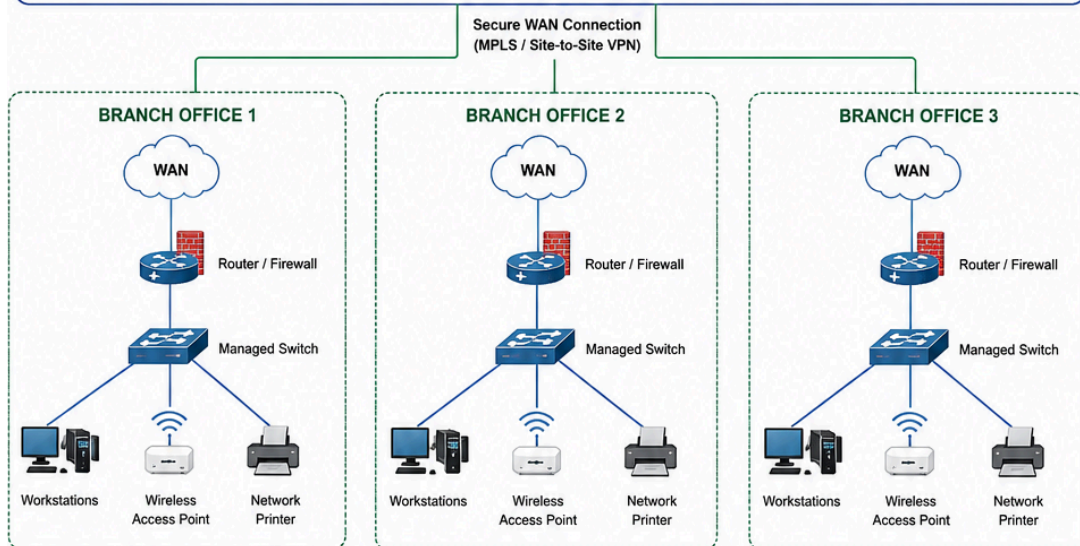
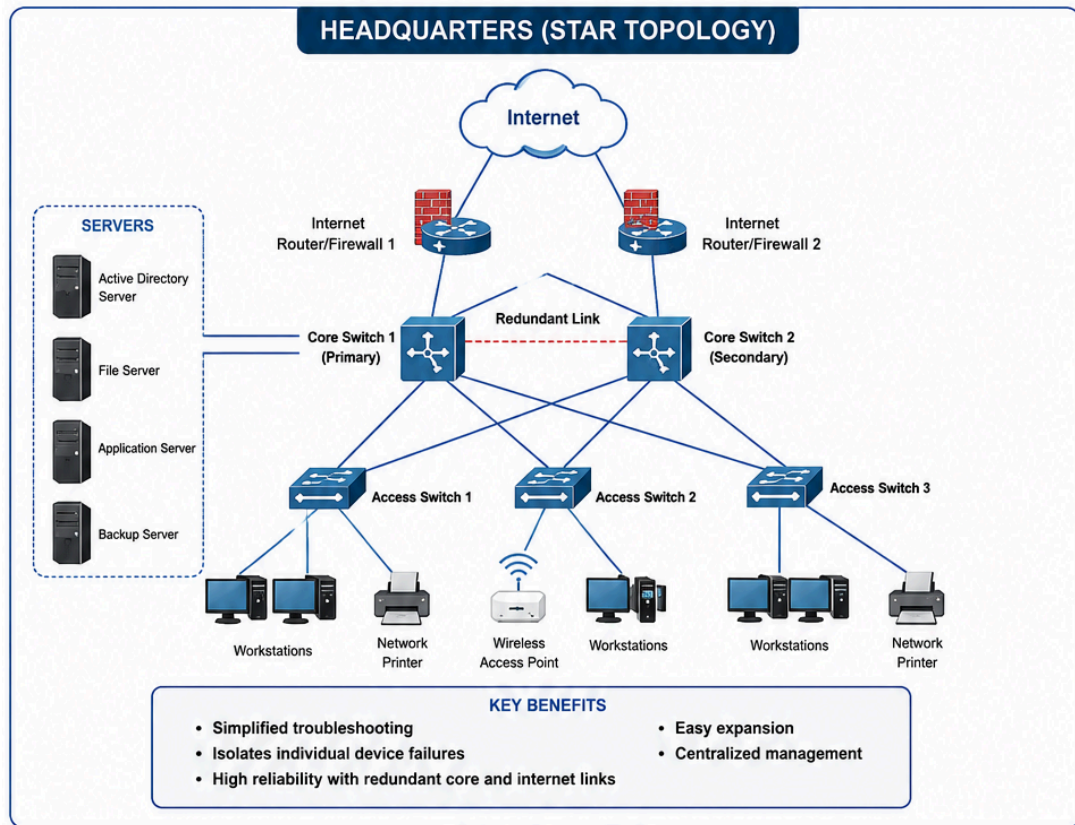
Branch Offices

Each **branch office** should also implement a **star topology** consisting of a router/firewall, managed switch, wireless access point, workstations, and network printer. Standardizing this design across all locations simplifies administration, maintenance, and future expansion.

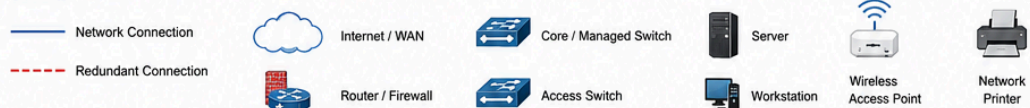
NETWORK TOPOLOGY

HEADQUARTERS AND BRANCH OFFICES

HEADQUARTERS (STAR TOPOLOGY)

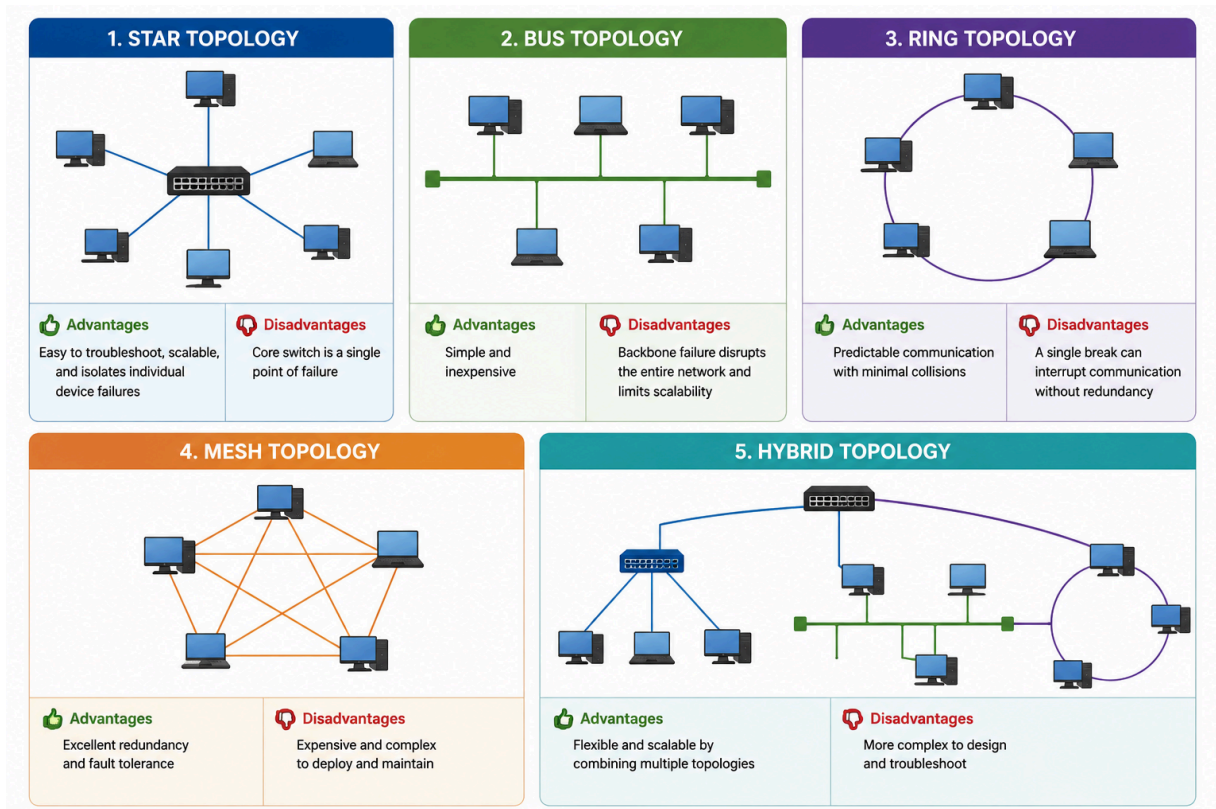


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Network Topology Comparison

Topology	Advantages	Disadvantages
Star	Easy to troubleshoot, scalable, and isolates individual device failures	Core switch is a single point of failure
Bus	Simple and inexpensive	Backbone failure disrupts the entire network and limits scalability
Ring	Predictable communication with minimal collisions	A single break can interrupt communication without redundancy
Mesh	Excellent redundancy and fault tolerance	Expensive and complex to deploy and maintain
Hybrid	Flexible and scalable by combining multiple topologies	More complex to design and troubleshoot



Because the company expects continued growth, the star topology provides the best balance of performance, reliability, and scalability.

Network Architecture Recommendation

A **client-server architecture** is recommended because it provides centralized management, security, authentication, and resource sharing. Unlike a peer-to-peer network, administrators can manage user accounts, permissions, software deployment, and security policies from centralized servers.

Core network services should include:

- Active Directory
- DHCP
- DNS
- File and Print Services
- Backup Services

Architecture Comparison

Architecture	Advantages	Disadvantages
Client-Server	Centralized authentication, security, management, and resource sharing	Server failures can affect many users without redundancy
Peer-to-Peer	Simple and inexpensive for small networks	Difficult to manage, secure, and scale

For a growing organization with multiple locations, a client-server architecture is the most secure and manageable solution.

Network Type Design

The proposed infrastructure combines several network types to meet business requirements.

Network Type	Description
LAN	Connects devices within a limited geographic area such as a home, school, or business.
WLAN	A wireless LAN providing network access using Wi-Fi.
WAN	Connects multiple LANs across large geographic distances.
MAN	Connects networks across a city or metropolitan area.
PAN	Connects personal devices over short distances using technologies such as Bluetooth.
VPN	Provides encrypted remote access for executives and authorized employees.

Together, these network types provide secure communication between offices while supporting remote users and future expansion.

IPv4 Addressing Scheme

Private IPv4 addressing using the **10.0.0.0/8** address space provides flexibility for future growth.

IPv4 Private Addressing

Location	IPv4 Network
Headquarters	10.1.0.0/16
Branch Office 1	10.2.0.0/16
Branch Office 2	10.3.0.0/16
Branch Office 3	10.4.0.0/16

Within headquarters, separate subnets should be assigned to departments such as Finance, Human Resources, Sales, Information Technology, Executive Offices, and Server Resources. Network segmentation reduces broadcast traffic, improves performance, and strengthens security.

IPv4 and IPv6 Network Services

To ensure compatibility with current systems while supporting future growth, the organization should implement both IPv4 and IPv6 using a dual-stack configuration.

IPv6 Addressing

Location	IPv6 Prefix
Headquarters	2001:db8:100::/64
Branch Office 1	2001:db8:200::/64
Branch Office 2	2001:db8:300::/64

Location	IPv6 Prefix
Branch Office 3	2001:db8:400::/64

Network Services

- Dynamic Host Configuration Protocol (DHCP) for automatic IPv4 address assignment
- DHCPv6 or Stateless Address Autoconfiguration (SLAAC) for IPv6 addressing
- Domain Name Server (DNS) for hostname resolution
- Network Time Protocol (NTP) for time synchronization
- Active Directory (AD) for centralized authentication and authorization

Implementing IPv4 and IPv6 in a dual-stack environment ensures compatibility with current technologies while preparing the organization for future networking requirements.

Implementation Plan

The network should be deployed in five phases to minimize disruption and verify successful implementation.

Phase	Activities
1. Planning	Gather requirements, design the network, develop addressing plans, and select hardware.
2. Infrastructure Deployment	Install routers, managed switches, servers, and wireless access points.
3. Network Configuration	Configure VLANs, DHCP, DNS, routing, IPv4/IPv6 services, and WAN connectivity.
4. Security Implementation	Configure firewalls, VPN, ACLs, IDS/IPS, and centralized monitoring.

Phase	Activities
5. Testing and Validation	Verify connectivity, routing, network services, VPN access, and overall performance.

Implementing the network in phases reduces deployment risks while ensuring each component functions correctly before moving to the next stage.

Security Considerations

Protecting organizational resources requires multiple layers of security to defend against both internal and external threats.

Recommended security controls include:

- Enterprise firewalls
- Intrusion Detection Systems (IDS)
- Intrusion Prevention Systems (IPS)
- Virtual Private Network (VPN) encryption
- Multi-Factor Authentication (MFA)
- VLAN segmentation and Access Control Lists (ACLs)
- Routine vulnerability assessments using Nmap
- Continuous monitoring using SNMP and Wireshark
- Security Information and Event Management (SIEM) for centralized log collection and event correlation
- Routine software updates and security patch management
- Regular backups and disaster recovery planning
- Security awareness training for employees
- House all critical networking equipment in secure, access-controlled equipment rooms with locked entry and access logging.

Implementing layered security controls strengthens the organization's ability to prevent, detect, investigate, and respond to cybersecurity threats while protecting critical business resources.

Conclusion

The proposed network design provides a secure, scalable, and reliable infrastructure capable of supporting the organization's current operations and future expansion. By combining star topologies, a client-server architecture, LAN/WLAN/WAN connectivity, VPN remote access,

structured IPv4 and IPv6 addressing, and centralized network services and management, the organization will establish a strong foundation for long-term growth.

Implementing layered security controls, proactive monitoring, and centralized management will improve network performance, simplify administration, protect organizational resources, and support future expansion with minimal disruption. This design meets the company's current requirements while providing the flexibility and resilience needed to adapt to evolving business and technology demands.

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